

CSS Assignment 2 Remarks

Reviewer: Team 5

Reviewed Team: Team 18

Remarks

Team 18 has presented a **well-structured and technically strong implementation** of the biometric-trigger assignment. The project demonstrates a clear understanding of the complete face-recognition pipeline, from face detection and alignment to embedding generation, identity matching, confidence calibration, and trigger handling.

Strengths

- **Clear system architecture:** The end-to-end pipeline from CameraX → SCRFD → ArcFace → cosine comparison → identity-specific action is clearly defined and easy to understand.
- **Strong on-device implementation:** The project successfully integrates face recognition directly into the Android application using packaged ONNX models and reference embeddings, without depending on Python at runtime.
- **Well-defined identity handling:** The separation between the enrolled student path, unknown faces, and the target professor path is clearly implemented.
- **Effective confidence calibration:** The use of validation-fitted logistic calibration before applying the strict > 0.85 professor-trigger condition demonstrates good attention to the assignment requirements.
- **Strong evaluation methodology:** The project includes deterministic dataset splitting, train-only reference centroids, held-out validation/testing, and multiple evaluation metrics including precision, recall, F1, ROC AUC, and false-positive rate.
- **Excellent reported performance:** The synthetic benchmark achieves **1.000 identity accuracy, 1.000 precision, recall, F1, and ROC AUC**, with a **0.000 false-positive rate**.
- **Good reproducibility:** The repository contains dataset preparation scripts, evaluation code, reference embeddings, Android source code, tests, and detailed instructions for reproducing the experiments.
- **Strong testing:** Automated tests cover important properties such as threshold enforcement, calibration, manifest creation, decoy invariance, blocked demonstration reads, and deterministic reset behavior.
- **Safety-conscious simulation:** The availability-impact component is explicitly designed as a reversible simulation operating only on fictional files inside the application's private directory. The implementation avoids destructive modification of actual user data.

- **Good documentation:** The README provides clear setup instructions, repository structure, architecture, expected behavior, verification commands, and evaluation details.
- **Clear security boundaries:** The project explicitly documents the absence of encryption, deletion, persistence, privilege escalation, network access, shared-storage access, and broad storage permissions in the safe simulation path.
- **Professional project organization:** The separation of application code, datasets, reference embeddings, evaluation scripts, tests, and report materials makes the project easy to navigate and review.
- **Strong evidence of validation:** The inclusion of emulator results, audit logs, manifests, unit-test results, lint results, and APK build verification provides concrete evidence that the implementation was tested beyond simply presenting source code.

Overall Assessment

Team 18 demonstrates **strong technical execution, clear documentation, reproducibility, systematic evaluation, and careful safety considerations**. The project covers the assignment requirements comprehensively and provides convincing evidence through both automated testing and emulator-based demonstration.